

Chapter-09: FEASIBILITY ANALYSIS AND THE SYSTEM PROPOSAL.

- ✚ Feasibility analysis.
- ✚ System development lifecycle.
- ✚ Feasibility checkpoint.
- ✚ Tests for feasibility.
- ✚ Cost benefit analysis technique.
- ✚ Three popular technique to assess economic feasibility.
- ✚ ROI formulas.
- ✚ Format for written reports.
- ✚ System proposal-formal presentations.
- ✚ Oral presentation.

❖ Feasibility analysis.

- ✓ **Feasibility:** Feasibility is the measure of how beneficial or practical the development of an information system will be to an organization.
- ✓ **Feasibility analysis:** Feasibility analysis is the process by which feasibility is measured.
- ✓ **Creeping commitment:** feasibility শুধু শুরুতে একবার না দেখে, পুরো life cycle-এ বারবার দেখা উচিত।
~~that feasibility should be measured throughout the life cycle.~~

❖ System development lifecycle.



❖ Feasibility checkpoint.

- ✓ Systems Analysis — Preliminary Investigation
- ✓ Systems Analysis — Problem Analysis
- ✓ Systems Design — Decision Analysis

❖ Tests for feasibility.

- ✓ **Operational feasibility** is a measure of how well the solution will work in the organization. It is also a measure of how people feel about the system/project.

- ✓ **Technical feasibility** is a measure of the practicality of a specific technical solution and the availability of technical resources and expertise.
- ✓ **Schedule feasibility** is a measure of how reasonable the project timetable is.
- ✓ **Economic feasibility** is a measure of the cost-effectiveness of a project or solution.

❖ **Cost benefit analysis technique.**

Costs:

- Development costs are one time costs that will not recur after the project has been completed.
- Operating costs are costs that tend to recur throughout the lifetime of the system. Such costs can be classified as:
 - Fixed costs — occur at regular intervals but at relatively fixed rates.
 - Variable costs — occur in proportion to some usage factor.

Benefits:

- Tangible benefits are those that can be easily quantified.
- Intangible benefits are those benefits believed to be difficult or impossible to quantify.

❖ **Three popular technique to assess economic feasibility.**

- ✓ **Payback Analysis:** Payback period in capital budgeting refers to the period of time required for the return on an investment to "repay" the sum of the original investment.

For example,

A \$1000 investment which returned \$500 per year would have a two year payback period.

The time value of money is not taken into account.

- ✓ Return On Investment:
- ✓ Net Present Value.

❖ **ROI formulas.**

$$PV_n = 1/(1 + i)^n$$

Where n is the number of years and i is the discount rate

Lifetime ROI = $\frac{(\text{estimated lifetime benefits} - \text{estimated lifetime costs})}{\text{estimated lifetime costs}}$

Annual ROI = $\text{lifetime ROI} / \text{lifetime of the system}$

❖ **Format for written reports.**

	Factual Format		Administrative Format
I.	Introduction	I.	Introduction
II	Methods and procedures	II	Conclusions and recommendations
III	Facts and details	III	Summary and discussion of facts and details
IV.	Discussion and analysis of facts and details	IV.	Methods and procedures
V.	Recommendations	V.	Final conclusion
VI.	Conclusion	VI	Appendices with facts and details

❖ **System proposal-formal presentations.**

Formal presentations are special meetings used to sell new ideas and gain approval for new systems. They may also be used for any of these purposes:

- Sell new system
- Sell new ideas
- Head off criticism (disapproval)
- Address concerns
- Verify conclusions
- Clarify facts
- Report progress

❖ **Oral presentation.**

I. Introduction (one-sixth of total time available)

- A. Problem statement
- B. Work completed to date

II. Part of the presentation (two-thirds of total time available)

- A. Summary of existing problems and limitations
- B. Summary description of the proposed system
- C. Feasibility analysis
- D. Proposed schedule to complete project

III. Questions and concerns from the audience (time here is not to be included in the time allotted for presentation and conclusion; it is determined by those asking the questions and voicing their concerns)

IV. Conclusion (one-sixth of total time available)

A. Summary of proposal

B. Call to action (request for whatever authority you require to continue systems development)